DO NOW — RECALL

Lesson 074 · Big O: Constant, Linear and Logarithmic

Name: _____ Class: _____ Date: _____

1. What is an algorithm's input size n?

2. What is a trace table?

3. What is an invariant?

4. A sorted search set of 128 candidates is halved repeatedly to one. How many halvings? Show the sequence.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Number of items or other measure of problem size.

2. Model answer

Record of intermediate variables/states.

3. Model answer

Property maintained through a loop.

4. Model answer

128→64→32→16→8→4→2→1; seven halvings.

By the end, you can…

• Big O describes the upper bound on how work grows with input size, not the precise milliseconds of an execution.

• A fixed direct index operation is O(1), scanning n values is O(n), and repeatedly halving n is O(log n).

• For n=64, a linear scan may inspect 64 items, while halving 64→1 takes six halvings. Sorting requirements and constants still matter.

Vocabulary — know these terms

O(1) — Constant asymptotic work.

O(n) — Linear growth.

O(log n) — Logarithmic growth.

asymptotic — Behaviour as size becomes large.

NEW LEARNING

Read, annotate and explain

Observe

Big O describes the upper bound on how work grows with input size, not the precise milliseconds of an execution.

Explain

A fixed direct index operation is O(1), scanning n values is O(n), and repeatedly halving n is O(log n).

Evaluate

For n=64, a linear scan may inspect 64 items, while halving 64→1 takes six halvings. Sorting requirements and constants still matter.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Classify loop performing one action for each item in n-element array.

Explain

The number of operations doubles when n doubles.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Match direct array access, full array scan and repeated halving with Big O. [6]

Question 2

For input size64, compare scan steps64 and repeated halving to1; calculate halvings. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why O(log n) binary search may be unsuitable for unsorted data even if asymptotically faster than linear search. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Big O directly tells you the elapsed seconds of a program.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.